Credit Risk Assessment System – BA Case Study

Project Background:

Banks require an automated system to evaluate loan applications and assess credit risk based on financial and credit history data. This project simulates the analysis and documentation required to build such a system.

Business Objective:

- Automate loan risk evaluation
- Reduce manual decision-making
- Improve loan approval efficiency
- Minimize credit default risk

Scope:

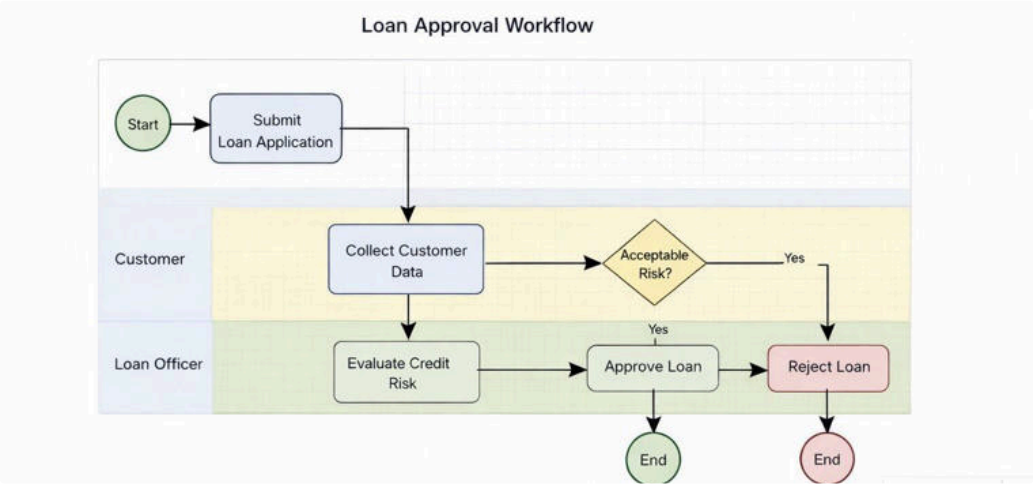
In Scope

- Loan application submission
- Risk score calculation
- Loan approval or rejection workflow

Out of Scope

- Actual credit bureau integrations
- Payment processing systems

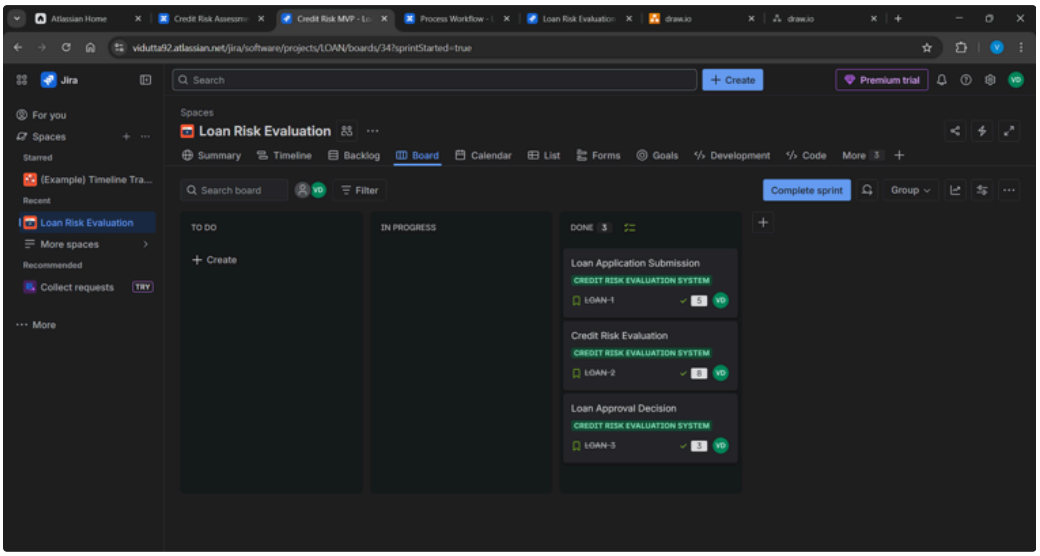
Process Flow:



Key Deliverables :

- Business Requirements Document (BRD) Credit Risk Assessment System
- Functional Requirements Specification (FRS) FRS- Credit Risk Assessment System
- User Stories & Epic (from Jira) LOAN-4: Credit Risk Evaluation System TO DO
- UML Use Case Diagram UML Use Case Diagram – Credit Risk Assessment System
- Requirement Traceability Matrix Requirement Traceability Matrix-Credit Risk Assessment System
- System Wireframes System Wireframes – Credit Risk Assessment System

Agile Delivery (Jira) :



- User stories were created and managed in Jira using a Scrum board.
- Stories were estimated using story points and grouped under an Epic representing the Credit Risk Assessment workflow.
- The stories progressed through the workflow stages from To Do → In Progress → Done during the sprint.

Tools Used :

- Confluence – Documentation & knowledge base
- Jira – User stories and backlog management
- Flowchart Maker & Online Diagram Software – BPMN, UML, and wireframes

Credit Risk Assessment System

Project Overview:

The Credit Risk Assessment System helps financial institutions evaluate loan applicants by analyzing customer financial data and predicting the probability of loan default. The system assists loan officers in making informed approval decisions while minimizing financial risk.

Business Objectives:

- Reduce loan default risk
- Improve accuracy of loan approval decisions
- Automate credit risk evaluation
- Provide data-driven insights for loan officers

Stakeholders

Stakeholders	Responsibility
Loan Officer	Approves/Rejects loan Applications
Risk Analyst	Evaluates customer credit risk
Data Analyst	Builds predictive models
IT Team	Implements the system

Problem Statement:

Financial institutions often face losses due to loan defaults caused by insufficient risk assessment during loan approval. The organization requires a system that can evaluate borrower risk using historical credit data.

Scope:

In Scope

- Customer loan application data analysis
- Credit risk prediction model
- Risk scoring dashboard
- Loan approval recommendation system

Out of Scope

- Loan disbursement processing
- Payment collection systems

Functional Requirement:

Index	Requirement
FR-1	System should allow loan officers to input customer financial data
FR-2	System should calculate credit risk score
FR-3	System should classify customers into risk categories
FR-4	System should display loan approval recommendation

Non-Functional Requirement:

- System response time < 3 seconds
- Secure storage of customer financial data
- System availability 99% uptime

Success Metrics:

- Reduction in loan default rate
- Improved loan approval decision accuracy
- Faster loan processing time

Assumptions:

- Customer financial data will be available during loan application.
- Historical credit data is available for risk model training.
- Loan officers will use the system dashboard to evaluate applications.

Constraints:

- Credit evaluation must complete within 3 seconds.
- System must comply with financial data security standards.
- Model predictions must be explainable for regulatory compliance.

FRS- Credit Risk Assessment System

Introduction:

Purpose

This document describes the functional requirements of the Credit Risk Assessment System. It defines system capabilities required to evaluate loan applicants and assist loan officers in decision-making.

Scope

The system will collect customer financial data, evaluate credit risk using predefined rules, and provide loan approval recommendations.

System Overview:

The Credit Risk Assessment System is designed to support financial institutions in assessing loan applicants by calculating credit risk scores based on applicant financial data.

Actors:

Actor	Description
Customer	Applies for loan
Loan Officer	Reviews loan applications
Risk Evaluation System	Calculates credit risk score
Data Analyst	Maintains risk model

Functional Requirements:

FR ID	Requirement Description	Priority	Related User Story
FR-1	System should allow customer to submit loan application	HIGH	Loan Application Submission
FR-2	System should collect customer financial data	HIGH	Loan Application Submission
FR-3	System should calculate credit risk score	HIGH	Credit Risk Evaluation
FR-4	System should classify applicants into risk categories	MEDIUM	Credit Risk Evaluation

FR-5	System should display risk score to loan officer	HIGH	Loan Approval Decision
FR-6	System should provide loan approval recommendation	MEDIUM	Loan Approval Decision

Each requirement is linked to user stories managed in Jira to support Agile development and traceability.

Business Rules:

Rule ID	Business Rule
BR-1	Applicants with high default probability should be flagged as high risk
BR-2	Risk score must be calculated below loan decision
BR-3	Loan officer can override system recommendation

Data Requirement:

Data Field	Description
Applicant_ID	Unique identifier
Credit_Score	Customer credit rating
Annual_Income	Applicant income
Loan_Amount	Requested loan amount
Default_Risk	Predicted default probability

Assumptions:

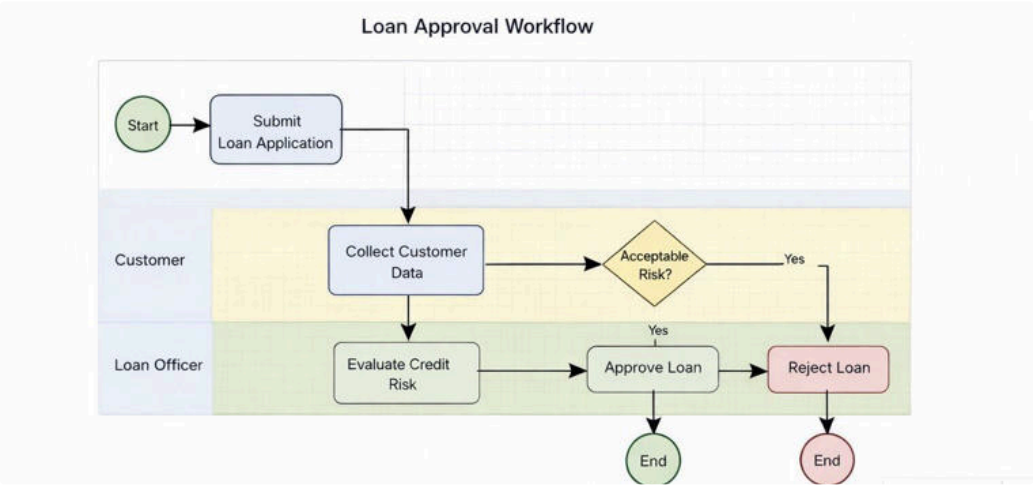
- Customer financial data is available during application.
- Historical data is available to evaluate credit risk.

Constraints:

- Risk evaluation must complete within 3 seconds
- System must comply with financial data security regulations

Process Workflow - Loan Approval

Loan Approval Workflow



This workflow describes the process of evaluating customer loan application using an automated credit risk scoring systems before the final approval decision.

Requirement Traceability Matrix-Credit Risk Assessment System

BRD ID	Functional Requirement	User Story	Epic	Status
BR-1	Loan Application Submission	Customer submits loan application	Credit Risk Evaluation System	Open
BR-2	Collect Customer Financial Data	Customer enters financial details	Credit Risk Evaluation System	Open
BR-3	Calculate Credit Risk Score	System evaluates credit risk	Credit Risk Evaluation System	Open
BR-4	Loan Approval Decision	Loan officer approves/rejects loan	Credit Risk Evaluation System	Open

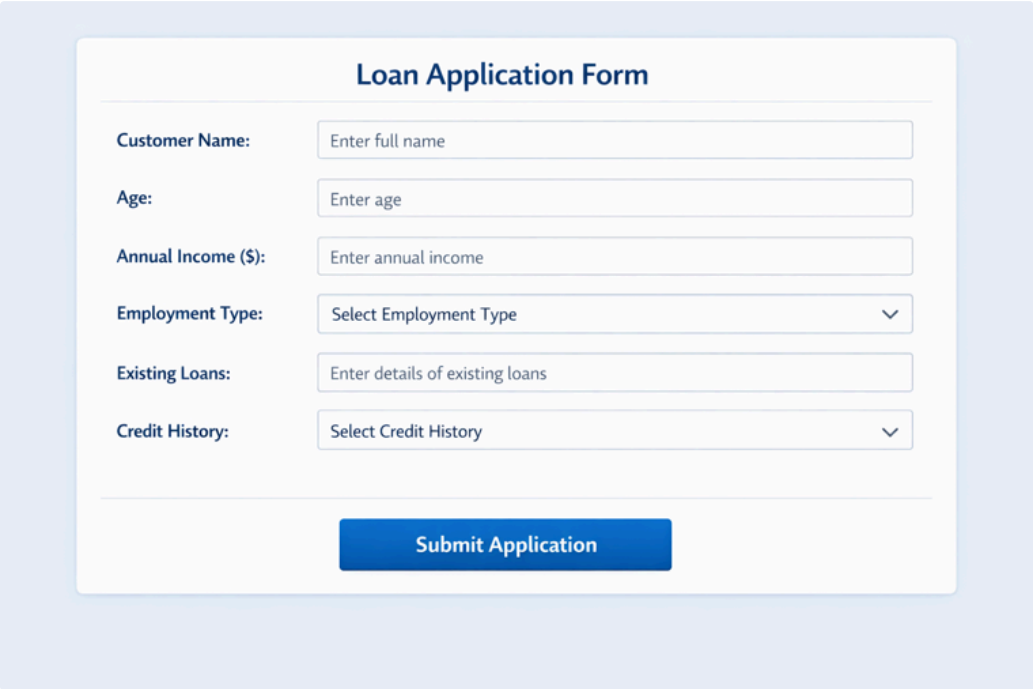
System Wireframes – Credit Risk Assessment System

📘 Overview:

This document presents the UI wireframes for the Credit Risk Assessment System. The wireframes illustrate the proposed layout and interaction for key screens involved in the loan application and risk evaluation process. These designs serve as a reference for developers and stakeholders during system implementation.

1. Loan Application Screen:

This screen allows loan applicants or bank officers to enter applicant details required for credit risk evaluation.



Fields:

Field Name	Field Type	Description
Customer Name	Text Field	Applicant full name
Age	Number Field	Applicant age
Annual Income	Number Field	Annual income of the applicant
Employment Type	Dropdown	Salaried / Self-Employed

Existing Loans	Number Field	Number of current loans
Credit History	Dropdown	Good / Average / Poor
Submit Application	Button	Submits application for risk evaluation

2. Risk Evaluation Dashboard:

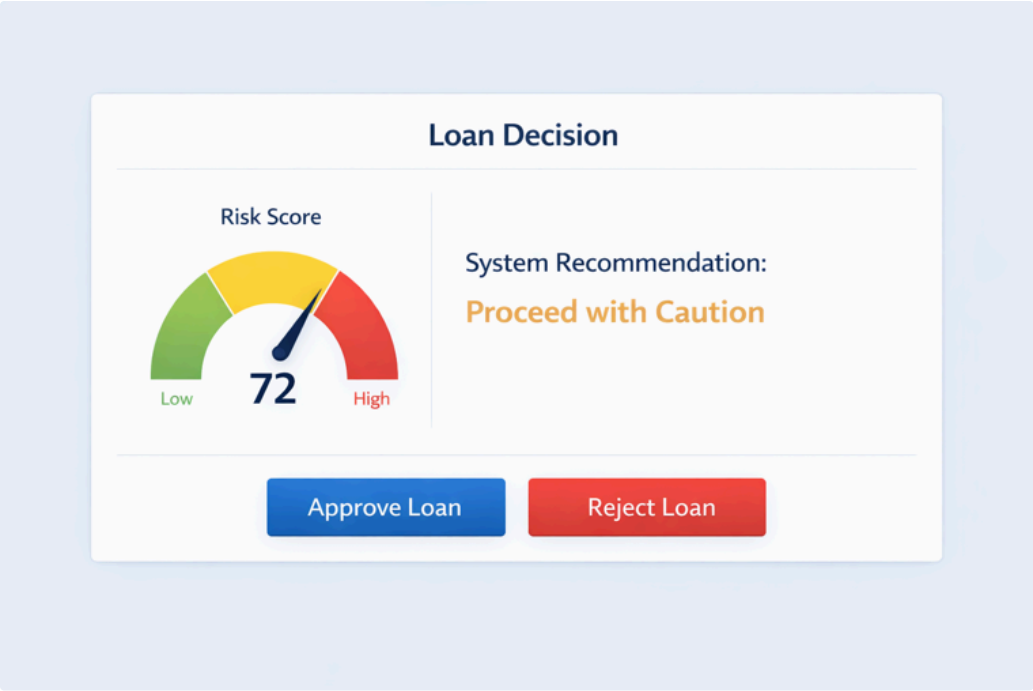
This screen allows loan officers to review the calculated credit risk score based on applicant financial information.

Elements:

Element	Description
Applicant Details	Displays submitted customer information
Risk Score	Numerical score generated by system
Risk Category	Low / Medium / High

3. Loan Decision Screen:

This screen allows loan officers to approve or reject the loan application based on the risk assessment.



Actions:

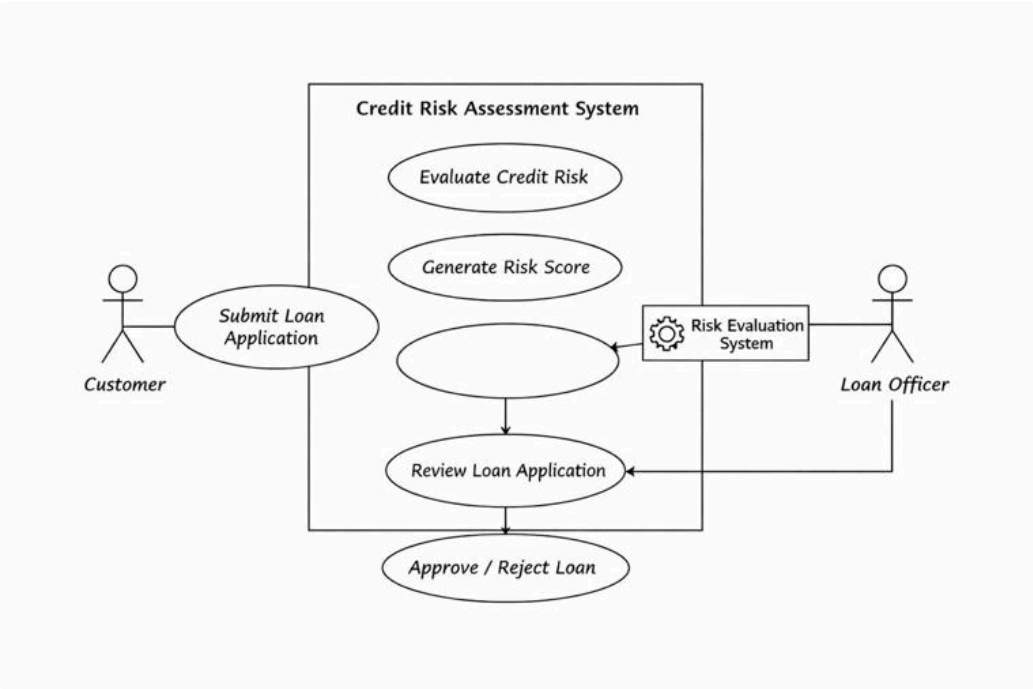
Action	Description
Approve Loan	Approves loan for applicant
Reject Loan	Rejects loan application

Note :

- Wireframes represent a conceptual UI layout.
- Final UI design may change during development.
- Risk score calculation is based on predefined credit evaluation rules.

UML Use Case Diagram – Credit Risk Assessment System

Diagram:



Description:

The Use Case Diagram illustrates how different actors interact with the Credit Risk Assessment System during the loan evaluation process. It highlights the main system functions involved in assessing credit risk and assisting loan approval decisions

Actors:

Actor	Description
Customer	Submits loan application and financial information
Loan Officer	Reviews application and makes approval decisions
Risk Evaluation System	Calculates credit risk score based on financial data

Use Cases:

Use Case	Description

Submit Loan Application	Customer submits financial and personal data
Evaluate Credit Risk	System analyzes financial data and risk factors
Generate Risk Score	System generates a credit score for the applicant
Review Loan Application	Loan officer reviews the application and risk score
Approve / Reject Loan	Loan officer decides the loan outcome

Business Value:

The UML diagram provides a visual understanding of system interactions, helping stakeholders understand how loan applications are processed and how risk evaluation supports decision making.

Assumptions:

- Customer financial data is accurate and validated
- Credit bureau data is accessible
- Loan officers have system access permissions

Dependencies:

- Integration with external credit bureau
- Availability of historical loan data
- Risk scoring model developed by Data Analytics team